

Jialin Ding

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jialinding.github.io

Revised 08/2026

EMPLOYMENT	Princeton University Assistant Professor, Computer Science	2025–Present
EDUCATION	Massachusetts Institute of Technology PhD, Computer Science Advisor: Tim Kraska	2018–2022
	Stanford University Bachelor of Science with Distinction, Electrical Engineering Minor in Economics	2014–2018
CONFERENCE PUBLICATIONS	C1. KramaBench: A Benchmark for AI Systems on Data-to-Insight Pipelines over Data Lakes. Eugenie Lai, Gerardo Vitagliano, Ziyu Zhang, Om Chabra, Sivaprasad Sudhir, Anna Zeng, Anton A. Zabreyko, Chenning Li, Ferdi Kossmann, Jialin Ding, Jun Chen, Markos Markakis, Matthew Russo, Weiyang Wang, Ziniu Wu, Michael J. Cafarella, Lei Cao, Samuel Madden and Tim Kraska. <i>ICLR 2026</i>. C2. Parachute: Single-Pass Bi-Directional Information Passing. Mihail Stoian, Andreas Zimmerer, Skander Krid, Amadou Latyr Ngom, Jialin Ding, Tim Kraska and Andreas Kipf. <i>VLDB 2025</i>. C3. Automated Multidimensional Data Layouts in Amazon Redshift. Jialin Ding, Matt Abrams, Sanghita Bandyopadhyay, Luciano Di Palma, Yanzhu Ji, Davide Pagano, Gopal Paliwal, Panos Parchas, Pascal Pfeil, Orestis Polychroniou, Gaurav Saxena, Aamer Shah, Amina Voloder, Sherry Xiao, Davis Zhang and Tim Kraska. <i>SIGMOD 2024 Industrial Track</i>. C4. SageDB: An Instance-Optimized Data Analytics System. Jialin Ding, Ryan Marcus, Andreas Kipf, Vikram Nathan, Aniruddha Nrusimha, Kapil Vaidya, Alexander van Renen and Tim Kraska. <i>VLDB 2023</i>. C5. APEX: A High-Performance Learned Index on Persistent Memory. Baotong Lu, Jialin Ding, Eric Lo, Umar Farooq Minhas and Tianzheng Wang. <i>VLDB 2022</i>. C6. Self-Organizing Data Containers. Samuel Madden, Jialin Ding, Tim Kraska, Sivaprasad Sudhir, David Cohen, Timothy Mattson and Nesime Tatbul. <i>CIDR 2022</i>. C7. Tsunami: A Learned Multi-dimensional Index for Correlated Data and Skewed Workloads. Jialin Ding, Vikram Nathan, Mohammad Alizadeh and Tim Kraska. <i>VLDB 2021</i>. C8. Instance-Optimized Data Layouts for Cloud Analytics Workloads. Jialin Ding, Umar Farooq Minhas, Badrish Chandramouli, Chi Wang, Yinan Li, Ying Li, Donald Kossmann, Johannes Gehrke and Tim Kraska. <i>SIGMOD 2021</i>. C9. ALEX: An Updatable Adaptive Learned Index. Jialin Ding, Umar Farooq Minhas, Jia Yu, Chi Wang, Jaeyoung Do, Hantian Zhang, Yinan Li, Badrish	

Chandramouli, Johannes Gehrke, Donald Kossmann, David Lomet and Tim Kraska. *SIGMOD 2020*.

- C10. **Learning Multi-dimensional Indexes.** Vikram Nathan*, Jialin Ding*, Mohammad Alizadeh and Tim Kraska. *SIGMOD 2020*.
- C11. **SageDB: A Learned Database System.** Tim Kraska, Mohammad Alizadeh, Alex Beutel, Ed Chi, Jialin Ding, Ani Kristo, Guillaume Leclerc, Samuel Madden, Hongzi Mao and Vikram Nathan. *CIDR 2019*.
- C12. **Moment-Based Quantile Sketches for Efficient High Cardinality Aggregation Queries.** Edward Gan, Jialin Ding, Kai Sheng Tai, Vatsal Sharan and Peter Bailis. *VLDB 2018*.

WORKSHOP PUBLICATIONS

- W1. **A Simple and Fast Way to Handle Semantic Errors in Transactions.** Jinghan Zeng, Eugene Wu, Sanjay Krishnan, Wyatt Lloyd and Jialin Ding. *Northeast AI Agents Day. Abstract*.
- W2. **TailorSQL: A NL2SQL System Tailored for Your Query Workload.** Kapil Vaidya, Jialin Ding, Sebastian Kosak, David Kernert, Chuan Lei, Xiao Qin, Abhinav Tripathy, Ramesh Balan, Balakrishnan Narayanaswamy and Tim Kraska. *AIDB Workshop @ VLDB 2025*.
- W3. **Utilizing Past User Feedback for More Accurate Text-to-SQL.** Matthias Urban, Jialin Ding, David Kernert, Kapil Vaidya and Tim Kraska. *HILDA Workshop @ SIGMOD 2025*.
- W4. **Learning Bit Allocations for Z-Order Layouts in Analytic Data Systems.** Jenny Gao, Jialin Ding, Sivaprasad Sudhir and Samuel Madden. *aiDM Workshop @ SIGMOD 2024*.
- W5. **Learning Bit Allocations for Z-Order Layouts in Analytic Data Systems.** Jenny Gao, Jialin Ding, Sivaprasad Sudhir and Samuel Madden. *Systems for ML Workshop @ NeurIPS 2023*.
- W6. **The Case for Learned Spatial Indexes.** Varun Pandey, Alexander van Renen, Andreas Kipf, Ibrahim Sabek, Jialin Ding and Alfons Kemper. *AIDB Workshop @ VLDB 2020*.
- W7. **LISA: Towards Learned DNA Sequence Search.** Darryl Ho, Jialin Ding, Sanchit Misra, Nesime Tatbul, Vikram Nathan, Vasimuddin Md and Tim Kraska. *Systems for ML Workshop @ NeurIPS 2019. Oral Presentation*.
- W8. **Learning Multi-dimensional Indexes.** Vikram Nathan*, Jialin Ding*, Mohammad Alizadeh and Tim Kraska. *ML for Systems Workshop @ NeurIPS 2019. Oral Presentation*.
- W9. **Efficient Mergeable Quantile Sketches using Moments.** Edward Gan, Jialin Ding and Peter Bailis. *SysML 2018. Extended Abstract*.
- W10. **A Machine-Compiled Database of Genome-Wide Association Studies.** Volodymyr Kuleshov, Jialin Ding, Braden Hancock, Alexander Ratner, Christopher Re, Serafim Batzoglou and Michael Snyder. *ISMB 2017. Short Paper*.

JOURNAL PUBLICATIONS

- J1. **A Machine-compiled Database of Genome-wide Association Studies.** Volodymyr Kuleshov, Jialin Ding, Christopher Vo, Braden Hancock, Alexander Ratner, Yang Li, Christopher R, Serafim Batzoglou and Michael Snyder. *Nature Communications 2019*.

J2. **MacroBase: Prioritizing Attention in Fast Data.** Firas Abuzaid, Peter Bailis, Jialin Ding, Edward Gan, Samuel Madden, Deepak Narayanan, Kexin Rong and Sahaana Suri. *TODS* 2018.

ARCHIVED PUBLICATIONS

- A1. **ODIN: A NL2SQL Recommender to Handle Schema Ambiguity.** Kapil Vaidya, Abishek Sankararaman, Jialin Ding, Chuan Lei, Xiao Qin, Balakrishnan Narayanaswamy and Tim Kraska. *CoRR* 2025.
- A2. **Cortex: Harnessing Correlations to Boost Query Performance.** Vikram Nathan, Jialin Ding, Tim Kraska and Mohammad Alizadeh. *CoRR* 2020.

TEACHING

- COS 418 (Distributed Systems): Fall 2025, Fall 2026
- COS 598I (Modern Data Systems): Spring 2026
- Teaching Assistant, 6.887 (Machine Learning for Systems): Fall 2021

STUDENTS

- Jinghan Zeng (co-advised with Wyatt Lloyd): 2025–Present
- Karan Tandon (co-advised with Ravi Netravali): 2025–Present
- Sairaj Voruganti: 2026–Present
- Nicholas Yap (MSE): 2025–Present
- Polly Ren (MSE): 2025–Present
- Anupta Argo (MSE): 2026–Present

SERVICE

- Reviewer, VLDB: 2025 (Distinguished Reviewer Award), 2026
- Reviewer, SIGMOD: 2026, 2027
- Reviewer, VLDB Demo Track: 2022-2023
- Reviewer, VLDB Journal: 2023
- Student Volunteer, VLDB 2021
- Reviewer, TKDE: 2020

INVITED TALKS

Instance-Optimization: Rethinking Database Design for the Next 1000X
National University of Singapore October 2024

Towards Practical Instance-Optimized Systems
Facebook/Meta February 2022

Learned Index Structures for Dynamic and Multi-Dimensional Data
University of Washington (NWDS Seminar) February 2021

Instance-optimized Indexing and Storage
Cornell University (DB Seminar) October 2020
LADSIOS Workshop @ VLDB August 2021
Stanford Systems Seminar March 2022

Learning Multi-dimensional Indexes
Boston University (MiDAS Seminar) April 2020
New England Database Day January 2020

FELLOWSHIPS AND AWARDS

- SIGMOD Jim Gray Doctoral Dissertation Award (Honorable Mention), 2024
- Meta PhD Fellowship, 2021–2023
- NSF Graduate Research Fellowship Program (Honorable Mention), 2018
- MIT Jacobs Presidential Fellowship, 2018

INDUSTRY EXPERIENCE

Amazon Scholar, Amazon 2025–Present

- Working on cloud data systems.

Applied Scientist, Amazon 2022–2025

- Conducted research on instance-optimized database systems as part of a research team embedded within AWS Redshift.

Research Intern, Microsoft Research, Redmond Summer 2020

- Led research on a data layout framework for cloud analytics services, with applications to Azure Synapse, resulting in a SIGMOD 2021 publication.

Research Intern, Microsoft Research, Redmond Summer 2018

- Led research on an updatable learned index structure, resulting in a SIGMOD 2020 publication.

Software Engineer Intern, Google Summer 2016

- As part of Google Safe Browsing, implemented a MapReduce pipeline to integrate Chrome browser incident data into the evaluation of user downloads.

Software Engineer Intern, Thumbtack Summer 2015

- Worked on SEO, automatic text generation, and recommendation systems.